

LUMBINI CITY COLLEGE
Drivertole, Rupandehi

Subject : Data Structures and Algorithms

Full marks 30

Semester : III , Paper : CACS 201

Pass marks 12

Long question answer [1*10 = 10]

3. Write an algorithm for the dynamic implementation of Stack and Queue.
4. Define TOH problem. Find the minimum sequences of move when number of discs = 4 in TOH.

Short question answer [5*4=20]

5. Discuss ADT. Discuss stack and queue as ADT.
6. Write an algorithm for infix to prefix conversion and evaluation.
7. What is asymptotic notations? Explain with example.
8. Construct a Binary Search Tree (BST) for the given sequence of numbers: {50, 30, 70, 20, 40, 60, 80}

Lumbini City College

Affiliated to Tribhuvan University

Subject: Web Technology

Faculty: BCA

Semester: III

Full Marks: 30

Pass Marks: 12

Time: 1.5hrs

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group A

Attempt any four questions

4x5=20

1) Write HTML tag to generate the following table.

	Average		Red eyes
	height	weight	
Males	1.9	0.003	40%
Females	1.7	0.002	43%

2) What is CSS? Why it is recommended to use CSS in HTML pages?

3) Write HTML tag to generate following output

First name:

Last name:

E-mail:

☐ Male

☐ Female

4) Create a XML code that can store the name, address, phone and website of each student. The address element can have multiple instances and has a type attribute that can be either permanent or temporary. Write its equivalent DTD.

5) Write short notes on following:

- CSS selectors

• DTD

Group C

Attempt any One questions

1x10=10

9) Differentiate between HTML tags and attributes. Write a HTML and CSS to display your CV on a web page.

10) Explain different types of tier technology with suitable diagram.

Best of Luck

Lumbini City College
Mid-Term Examination 2081

Level: Bachelor
Programme: BCA

Internal Assessment

Course: System Analysis and Design

Year : 2081
Full Marks: 30
Time: 1.5hrs.

Attempt any Five questions.

1. What is Information System? Explain about different types of Information System. [6]
2. What are the phases of Waterfall Model? Explain. Also mention its advantages and disadvantages. [6]
3. Explain about the different stages of Agile Modelling. What are its advantages and disadvantages? [6]
4. What is Project Management? What are the phases of Project Management? Explain. [6]
5. Explain the three primary activities of identifying and selecting IS development projects. [6]
6. Describe the steps involved in corporate strategic planning. What are the three competitive strategies? [6]
7. What are the traditional methods for determining requirements? Explain in brief. [6]
8. What do you mean by feasibility analysis? Discuss about different categories of feasibility. [6]

Lumbini City College

Mid Term Examination - 2025

BCA/Third Semester: OOP in java

Full Marks: 30

Time: 1.5 Hrs.

Candidates are required to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt any five questions. Question no SIX is compulsory

[6*5= 30]

1. Define OOP. Explain features of Object Oriented Programming Language with suitable example.
2. Explain different types of control statements used in java.
3. Define method overriding? Write any program to implement concept of dynamic method dispatch.
4. Why it is important to handle exception in java? Write a program to illustrate the use of custom exception handling.
5. Define package. Explain Different access specifiers in java with suitable examples.
6. Write a program in java to check whether the given string is palindrome or not without using any in built library function.

GROUP A (Objective questions)

10 x1 =10

- i. In which measure all the observations are taken into consideration?
(a) Mean (b) Median (c) Mode (d) Partition Values
- ii. What can be determined graphically?
(a) Mean (b) Mode (c) Standard deviation (d) None
- iii. When inverse relationship exists between two variables then what is correlation between these variables?
(a) Positive (b) Negative (c) No (d) None
- iv. What is variance of observations 5,5,5,5,5?
(a) 0 (b) 5 (c) 25 (d) 4
- v. When values of one variable increase then values of other variable also increases and vice versa then what type of correlation exists?
(a) Positive (b) Negative (c) Auto (d) None
- vi. What is the meaning of b in regression equation $y = a + bx$?
(a) Change in x per unit change in y
(b) Change in y per unit change in x
(c) Y intercept
(d) Dependent variable
- vii. What type of sampling are cluster sampling, stratified sampling and systematic sampling?
(a) Direct sampling
(b) Indirect sampling
(c) Random sampling
(d) Non-random sampling
- viii. What is the range of probability?
(a) -1 to 1 (b) -1 to 0 (c) 0 to 1 (d) none
- ix. Which distribution has equal mean and variance?
(a) Binomial (b) Poisson (c) Normal (d) None
- x. What type of sampling is used when population is homogeneous?
(a) Simple random sampling
(b) Stratified sampling
(c) Systematic sampling
(d) Cluster sampling

10/11

GROUP B: Short questions Attempt all the questions (5 x 4 = 20)

1. Define measure of dispersion? What are the different types of measure of dispersion? Which is the best measure of dispersion? Explain with reasons.

2. Calculate the Coefficient of variation from the following data:

Age(year)	10-20	20-30	30-40	40-50	50-60
No. of person	8	20	15	17	25

3. The marks distribution of 100 students of a college is given below:

Marks	10-20	20-30	30-40	40-50	50-60
No of students	15	20	30	20	15

- Find the highest mark of the weakest 30% of the students.
 - Find the lowest mark of top 40% of the students
4. About twenty percent of users do not close windows properly. Suppose that windows are installed in a computer laboratory of a campus that is used by random students in a random order.
- What is the probability that exactly 8 of the next 10 students will close Windows properly?
 - What is the probability that at least 2 students will not close Windows properly out of 7 students?

5. A city council has gathered data on number of minor traffic accidents and the number of youth games that occurred in town over the weekends.

Football games	27	45	41	19	35	39	19
Minor accidents	57	64	80	46	62	72	52

Find the correlation coefficient between football games and minor accidents.

$$\frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$